

MICRO INVERTER

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A simple low-power inverter circuit is described here, which converts 12V DC into 230V AC. It can be used to power very light loads like night lamps and cordless telephones, but can be modified into a powerful inverter by adding more MOSFETs. This circuit has two stages—battery charger with cut-off, and battery level indicator and inverter circuit.

Charging circuit is built around IC1 (LM317) as shown in Fig. 1. When mains 230V AC is available, IC1 pro-

vides gate voltage to SCR1 (TYN616) through diode D3 (1N4007). SCR1 starts charging the battery. For output voltage setting preset VR1 may be used.

The battery level indicator and inverter circuit is shown in Fig. 2. The battery level checking system is built around transistors T1 and T2 (both BC547) along with some discrete components. When the battery is charged (say, to more than 10.50V), LED1 glows and piezo-buzzer PZ1 does not sound. On the other hand, when battery voltage goes down (say, below 10.50V), LED1 stops

glowing and piezo-buzzer sounds, indicating that the battery has been discharged and needs recharging for further use.

The inverter is built around IC2 (CD4047), which is wired as an astable multivibrator operating at a frequency of around 50 Hz. The Q and

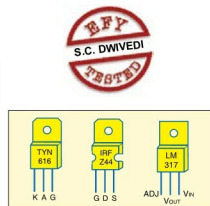


Fig. 3: Pin configurations of TYN616, IRFZ44 and LM317

$\bar{Q}$ outputs of IC2 directly drive power MOSFETs (T3 and T4). The two MOSFETs (IRFZ44) are used in push-pull configuration. The inverter output is filtered by capacitor C1.

Assemble the circuit on a general-purpose PCB and enclose it in a suitable metal box. Refer Fig. 3 for pin configurations before mounting the components on the PCB. Mount the transformer on the chassis and the battery in the box using supporting clamps. Use suitable heat-sinks for MOSFETs. The circuit can be used for other applications as well by delivering higher power with the help of a higher current rating transformer and additional MOSFETs. ●

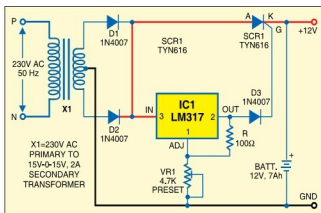


Fig. 1: Battery charger with cut-off circuit

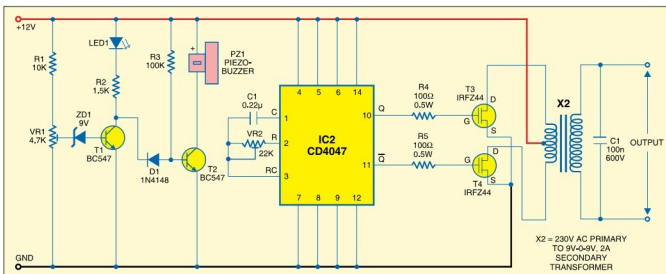


Fig. 2: Battery level indicator and inverter circuit